

| COURSE DESCRIPTION         |                                      |               |    |
|----------------------------|--------------------------------------|---------------|----|
| Program                    | Chemical Engineering                 |               |    |
| Course Title               | Fundamentals of Chemical Engineering |               |    |
| Course Code                | 2072                                 | Semester      | II |
| Type of Course             | Theory                               | Contact Hours | 60 |
| Teaching Scheme (L: T:P:R) | 4:0:0:0                              | Credits       | 4  |
| CIE Marks                  | 40                                   | ESE Marks     | 60 |

### Introduction

This foundational course traces the historical evolution and scope of chemical engineering while outlining the professional duties of chemical engineers. Students master essential unit conversions and basic calculations, gain an understanding of momentum, heat, and mass transfer operations, distinguish between unit operations and processes, and explore the fundamentals of electrical engineering and mechanics.

### Course Objectives

|   |                                                                                                                                                                 |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | To familiarise with the historical background of Chemical Engineering, including its scope and the professional duties of Chemical Engineering technicians.     |
| 2 | To impart knowledge on the basic calculations for unit conversions in chemical engineering contexts.                                                            |
| 3 | To familiarise with the basic principles of momentum, heat and mass transfer operations, along with the distinctions between unit operations and unit processes |
| 4 | To familiarise with the basic principles of electrical and electronics engineering relevant to chemical engineering.                                            |
| 5 | To familiarise with the basics of mechanics applicable to chemical engineering.                                                                                 |

### Course Pre-requisite

| Topic                       | Course code | Course Title                                                        | Semester |
|-----------------------------|-------------|---------------------------------------------------------------------|----------|
| Fundamentals of Physics     | 2002B       | Engineering Physics for Applied Electrical Technology and Computing | I        |
| Fundamentals of Mathematics | 1002        | Fundamentals of Engineering Mathematics                             | I        |

### Course Outcome

| CO  | Course Outcome                                                                                                                                                                          | Blooms Taxonomy Level |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | Summarise the historical background of chemical engineering, its scope, and the duties of a chemical engineering technician, while applying the basic calculations in unit conversions. | Understand            |
| CO2 | Summarise the fundamentals of unit operations, unit processes, momentum, heat, and mass transfer operations.                                                                            | Understand            |
| CO3 | Outline the basic principles and terminology of electrical and electronics engineering relevant to chemical processes.                                                                  | Understand            |
| CO4 | Summarise the fundamental concepts of mechanics applicable to chemical engineering.                                                                                                     | Understand            |

### CO-PO mapping

| COURSE ARTICULATION MATRIX |                  |     |     |     |     |     |     |     |     |      |      |
|----------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| Course Outcomes            | PROGRAM OUTCOMES |     |     |     |     |     |     |     |     |      |      |
|                            | PO1              | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
| CO1                        | 3                | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |
| CO2                        | 2                | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |
| CO3                        | 2                | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |
| CO4                        | 2                | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |
| Total                      | 9                | -   | -   | -   | -   | -   | -   | -   | -   | -    | 12   |
| Correlation Level          | 2.33             | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

### Teaching and Assessment Scheme

| Teaching Scheme/Week |   |   |   | Self-learning (SS)/Week | Total Hours/Semester | Credits C | Examination Scheme |     |       |           |     |       |       |
|----------------------|---|---|---|-------------------------|----------------------|-----------|--------------------|-----|-------|-----------|-----|-------|-------|
|                      |   |   |   |                         |                      |           | Theory             |     |       | Practical |     |       | Total |
| L                    | T | P | S |                         |                      |           | CIE                | ESE | Total | CIE       | ESE | Total |       |
| 4                    | 0 | 0 | 0 | 6                       | 60                   | 4         | 40                 | 60  | 100   | 0         | 0   | 0     | 100   |

*L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination*

### Syllabus - Major Topics

| Module     | Title                                        | Major Topics                                                                                                                                                                                                                                                                                                                                                                                                 | Instructional Hours |   |
|------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---|
|            |                                              |                                                                                                                                                                                                                                                                                                                                                                                                              | L                   | T |
| Module I   | Introduction to chemical industries          | Historical background and scope of Chemical Engineering - Duties of a Chemical Engineering Technician - Differentiate the duties of a chemical engineering technician and a chemist.<br>Modern Chemical industries - Batch and continuous processes - transition from batch to continuous processes.<br>Units and conversions - basic calculations.                                                          | 15                  | 0 |
| Module II  | Unit operations and unit processes           | Fundamentals of momentum, heat and mass transfer - Unit Operations - Unit Processes.                                                                                                                                                                                                                                                                                                                         | 15                  | 0 |
| Module III | Basic Electrical and Electronics Engineering | Current - Voltage - Resistance - Power - Ohm's law - Laws of resistance - Combination of resistances - series, parallel - DC circuit - AC circuit - Faraday's laws of electromagnetic induction - Generation of AC voltage - Transformers - Electric motors - Applications in chemical plants.<br>Electronic components - Resistors - Capacitors - Inductors - Diodes - Applications in chemical industries. | 15                  | 0 |
| Module IV  | Basics in Mechanics                          | Basics of mechanics - Statics - Dynamics - force - Resolution of force - Friction types and laws of friction - Stresses and strain - Types of stresses - Standard stress-strain curve - Mechanical properties of materials - Elasticity, stiffness, plasticity, toughness, brittleness, ductility, malleability and hardness.                                                                                | 15                  | 0 |

### Detailed Syllabus

| Subtopic                                    | Student Learning Outcome                                                                                                                                                                                                                                                 | Mode of delivery (L/T) | Mapped CO | Taxonomy Level | Instructional Hours |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------|----------------|---------------------|
| Module I                                    |                                                                                                                                                                                                                                                                          |                        |           |                |                     |
| Chemical engineering                        | Explain the fundamental principles of Chemical Engineering and its scope in process industries.                                                                                                                                                                          | L                      | CO1       | Understand     | 0.5                 |
| Importance of the chemical industry         | Explain the importance of the chemical industry in economic and industrial development and its significance in day-to-day life.                                                                                                                                          | L                      | CO1       | Understand     | 0.5                 |
| Types of industry                           | Classify chemical industries based on product type and scale of operation.<br>Identify major chemical industries in India and their key products.                                                                                                                        | L                      | CO1       | Understand     | 1                   |
| Industries in Kerala                        | Identify the raw materials and products of major chemical industries in Kerala.<br>Classify the major chemical industries in Kerala into categories such as fertiliser, petrochemical, petroleum refining, chlor-alkali, metallurgical, and organic chemical industries. | L                      | CO1       | Understand     | 1                   |
| Manpower in the chemical industry           | Explain the role and responsibilities of manpower in industrial operations and interpret a typical organisational chart of a process industry.                                                                                                                           | L                      | CO1       | Understand     | 1                   |
| Duties of a chemical engineering technician | Discuss the Duties of a Chemical technician<br>Differentiate the duties of a chemical engineering technician and a chemist                                                                                                                                               | L                      | CO1       | Understand     | 1                   |
| Modern chemical industries                  | Define Batch and continuous processes.<br>Discuss the need for transition from batch to continuous process                                                                                                                                                               | L                      | CO1       | Understand     | 1                   |

|                                                  |                                                                                                                                                                                                                                                           |   |     |            |   |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|------------|---|
| Process development in industry                  | Outline the stages of chemical process development and describe the role of a chemical engineering technician at each stage.<br>Outline laboratory scale, pilot plant scale, and commercial scale-up of process plant                                     | L | CO1 | Understand | 1 |
| Systems of unit                                  | Explain the FPS, MKS and SI systems of units.<br>Differentiate between fundamental quantities and derived quantities.<br>Describe the fundamental units in the SI system.                                                                                 | L | CO1 | Understand | 1 |
| Different fundamental units and their conversion | Identify the conversion of fundamental quantities, length, mass, time, and temperature, and derived quantities, volume, density, force, pressure, work/energy, heat, heat capacity, volumetric flow rate, mass flow rate, and absolute pressure.          | L | CO1 | Understand | 1 |
| Conversion of units                              | Convert from one system of units to another.<br>Conversion of force (kgf, lbf in to N)<br>Conversion of pressure (Pascal, torr, mmHg, atm, bar, kgf/cm <sup>2</sup> , etc. into N/m <sup>2</sup> )                                                        | L | CO1 | Apply      | 1 |
| Conversion of units                              | Conversion of Energy, Work, Heat (cal, Btu, kW.h in to J)<br>Conversion of Power (hp, Btu/h, kcal/h in to J/s)<br>Conversion of temperature from i) Celsius scale to Kelvin scale ii) Celsius to Fahrenheit scale iii) Fahrenheit scale to Rankine scale. | L | CO1 | Apply      | 1 |
| Mole concept                                     | Describe atomic weight, Molecular weight,                                                                                                                                                                                                                 | L | CO1 | Apply      | 1 |
| Mole concept                                     | Describe equivalent weight and moles.                                                                                                                                                                                                                     | L | CO1 | Apply      | 1 |
| Mole concept                                     | Calculate the molecular weight and the number of moles of mixtures                                                                                                                                                                                        | L | CO1 | Apply      | 1 |
| Mole concept                                     | Calculate the equivalent weight of compounds                                                                                                                                                                                                              | L | CO1 | Apply      | 1 |
| Module II                                        |                                                                                                                                                                                                                                                           |   |     |            |   |
| Fundamentals of momentum transfer                | Define momentum transfer, driving force for momentum transfer, and the basic classification of fluid mechanics                                                                                                                                            | L | CO2 | Understand | 1 |
| Fundamentals of heat transfer                    | Define heat transfer, and identify the driving force for heat transfer.<br>Identify different modes of heat transfer with examples                                                                                                                        | L | CO2 | Understand | 1 |
| Fundamentals of mass transfer                    | Define mass transfer.<br>Describe the driving force for mass transfer.                                                                                                                                                                                    | L | CO2 | Understand | 1 |
| Unit operations                                  | Differentiate between unit operations and unit processes.<br>Identify various unit operations and processes.                                                                                                                                              | L | CO2 | Understand | 1 |
| Unit operations: size reduction                  | Describe the mechanical operation of size reduction.                                                                                                                                                                                                      | L | CO2 | Understand | 1 |
| Unit operations: Size separation                 | Describe size separation, filtration, sedimentation, centrifugation and mixing                                                                                                                                                                            | L | CO2 | Understand | 1 |
| Unit operations: Mass transfer                   | Outline gas absorption, adsorption, drying, and distillation with a block diagram                                                                                                                                                                         | L | CO2 | Understand | 1 |
| Unit operations: Mass transfer                   | Outline crystallisation, extraction and leaching with a block diagram                                                                                                                                                                                     | L | CO2 | Understand | 1 |
| Unit operation in Heat transfer                  | Explain the evaporation operation                                                                                                                                                                                                                         | L | CO2 | Understand | 1 |
| Unit processes                                   | Explain unit processes: sulphonation, polymerisation, oxidation, and reduction, with examples                                                                                                                                                             | L | CO2 | Understand | 2 |
| Unit processes                                   | Explain unit processes: hydrogenation, hydration, saponification, alkylation, with examples                                                                                                                                                               | L | CO2 | Understand | 2 |
| Unit processes                                   | Explain the unit processes of esterification, nitration, chlorination, calcination, cracking, and pyrolysis with examples.                                                                                                                                | L | CO2 | Understand | 2 |
| Module III                                       |                                                                                                                                                                                                                                                           |   |     |            |   |

|                                                           |                                                                                                                                                                                                 |   |     |            |     |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|------------|-----|
| Electric current                                          | Illustrate the idea of electric potential using a simple voltaic cell.<br>Define Electric current, Resistance, Potential difference, resistivity, conductance and conductivity with their units | L | CO3 | Understand | 1   |
| Ohm's Law with application                                | State and explain Ohm's Law.<br>Explain resistance in series and parallel, and the voltage divider rule.                                                                                        | L | CO3 | Understand | 1   |
| Simple problems based on Ohm's Law                        | Solve simple problems to calculate the effective resistance in series and parallel circuits.                                                                                                    | L | CO3 | Apply      | 1   |
| AC and DC Circuits                                        | Explain DC and AC electric circuits                                                                                                                                                             | L | CO3 | Understand | 1   |
| Electromagnetic induction                                 | State Coulomb's Law with a mathematical expression<br>Define electromagnetism.                                                                                                                  | L | CO3 | Understand | 1   |
| Electromagnetic induction                                 | Identify Fleming's left-hand rule.<br>Identify the relation between magnetism and electricity.<br>Define induced emf and current.                                                               | L | CO3 | Understand | 1   |
| Faraday's Law                                             | State and explain Faraday's Laws of electromagnetic induction                                                                                                                                   | L | CO3 | Understand | 1   |
| Faraday's Law                                             | Identify Fleming's right-hand rule.                                                                                                                                                             | L | CO3 | Understand | 1   |
| Alternating current fundamentals                          | Discuss the generation of alternating voltage and current                                                                                                                                       | L | CO3 | Understand | 1   |
| Electric motors                                           | Illustrate the principle and working of an electric generator.                                                                                                                                  | L | CO3 | Understand | 1   |
| Electric motors                                           | Illustrate the principle and working of DC motor.                                                                                                                                               | L | CO3 | Understand | 0.5 |
| Transformers                                              | Explain the principle in a transformer.<br>Discuss the step-up transformers and step-down transformers.                                                                                         | L | CO3 | Understand | 0.5 |
| Electrical and electronic applications in Chemical plants | Discuss the major Applications of electrical equipment in chemical process plants                                                                                                               | L | CO3 | Understand | 1   |
| Electronic components                                     | Identify different resistors.<br>Identify different types of capacitors, inductors, and diodes.                                                                                                 | L | CO3 | Understand | 2   |
| Electronic application in the chemical Industry           | Discuss the major applications of electronic equipment in the chemical process plants                                                                                                           | L | CO3 | Understand | 1   |
| Module IV                                                 |                                                                                                                                                                                                 |   |     |            |     |
| Basics of mechanics                                       | Define 'engineering mechanics'.<br>Classify branches of mechanics                                                                                                                               | L | CO4 | Understand | 1   |
| Parallelogram law                                         | The gravitational law of attraction, the parallelogram law.                                                                                                                                     | L | CO4 | Understand | 1   |
| Resolution of force                                       | Discuss the composition of forces.<br>Define resultant force.                                                                                                                                   | L | CO4 | Understand | 1   |
| Resolution of force                                       | State the triangle law of force.<br>State polygon law of forces                                                                                                                                 | L | CO4 | Understand | 1   |
| Resolution of force                                       | Discuss the resolution of force with examples                                                                                                                                                   | L | CO4 | Understand | 1   |
| Laws of friction                                          | Identify different types of frictional forces.<br>Discuss the principles of the laws of friction.                                                                                               | L | CO4 | Understand | 2   |
| Stress and strain of materials                            | Define stress and strain with their units.<br>Identify the types of stress.                                                                                                                     | L | CO4 | Understand | 1   |
| Stress and strain of materials                            | Define tensile stress, compressive stress, and shear stress with mathematical expressions.                                                                                                      | L | CO4 | Understand | 2   |
| Modulus of elasticity and stress-strain curve             | Define modulus of elasticity.<br>Illustrate the general behaviour of the stress-strain curve of the material.                                                                                   | L | CO4 | Understand | 2   |
| Mechanical properties of materials                        | Define elasticity and stiffness.                                                                                                                                                                | L | CO4 | Understand | 1   |
| Mechanical properties of materials                        | Define plasticity, toughness, and brittleness.                                                                                                                                                  | L | CO4 | Understand | 1   |

|                                    |                                               |   |     |            |   |
|------------------------------------|-----------------------------------------------|---|-----|------------|---|
| Mechanical properties of materials | Explain ductility, malleability and hardness. | L | CO4 | Understand | 1 |
|------------------------------------|-----------------------------------------------|---|-----|------------|---|

### Suggested Student Activities for Self-Learning

| Week                      | Self-Learning description                                                                                                                                                                                                                                                                                         | Hours |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Module I                  |                                                                                                                                                                                                                                                                                                                   |       |
| 1                         | Prepare notes on the scope of chemical engineering. Visit the websites of various industries in India to learn about their major products.<br>Prepare notes to identify the duties of a chemical engineer                                                                                                         | 6     |
| 2                         | Visit the NPTEL courses on important steps in process development to understand the chemical process development.<br>Make a comparison table of Chemical Engineer/Technician/Chemist duties.<br>Sketch batch vs continuous process flow diagrams.<br>Refer to the textbook to identify the different unit systems | 6     |
| 3                         | Recall the mole concepts and definitions from the basic chemistry textbooks.<br>Solve simple numericals based on molecular weights, equivalent weights, and the number of moles of the mixture.                                                                                                                   | 6     |
| 4                         | Create a Chemical Engineering timeline chart (1850-present).<br>Prepare study notes for module I                                                                                                                                                                                                                  | 6     |
| Module II                 |                                                                                                                                                                                                                                                                                                                   |       |
| 5                         | Refer to textbooks on the fundamentals of chemical engineering to understand the basics of fluid mechanics, heat transfer, and mass transfer.<br>Prepare study notes on Fluid Mechanics, Heat Transfer, and Mass Transfer – Basic Concepts                                                                        | 6     |
| 6                         | Refer to text book on the concept of unit operations and unit processes.<br>Prepare notes for various unit operations taught that week                                                                                                                                                                            | 6     |
| 7                         | Refer to text book on various unit operations.<br>Prepare notes for various unit operations taught in that week.<br>Draw simple blocks of unit operations showing inputs and outputs                                                                                                                              | 6     |
| 8                         | Refer to text book on various unit processes.<br>Prepare notes for the various unit processes taught that week.<br>Prepare a table containing Process name → Type of reaction → Example industry/product.<br>Write an assignment covering CO-1 and CO2                                                            | 6     |
| Module III                |                                                                                                                                                                                                                                                                                                                   |       |
| 9                         | Refer to the textbook and prepare notes for the basics of electric current and Ohm's law.<br>Practice numerical thoughts that week.<br>Preparation for Series test 1 covering CO1 and CO2                                                                                                                         | 6     |
| 10                        | Refer to the textbook and prepare notes for electromagnetic induction and related topics taught that week.<br>Prepare a short note on "Applications of Electromagnetic Induction in daily life."                                                                                                                  | 6     |
| 11                        | Refer text book and prepare notes for alternating currents and electric motors.<br>Watch a short educational video on generator/motor working.                                                                                                                                                                    | 6     |
| 12                        | Prepare notes for basic electronic components.<br>Prepare a short note on "Applications of Electronics in Process Plant Operations".                                                                                                                                                                              | 6     |
| Module IV                 |                                                                                                                                                                                                                                                                                                                   |       |
| 13                        | Refer to the textbook and prepare notes for the basics of mechanics and Newton's law.<br>Prepare a poster illustrating Newton's law.                                                                                                                                                                              | 6     |
| 14                        | Solve numericals for the resolution of forces.<br>Prepare the notes for the laws of friction.<br>Refer to the textbook and prepare notes on the stress and strain of materials.<br>Observe the stress-strain curves for various materials.                                                                        | 6     |
| 15                        | Refer to the textbook and prepare notes for the mechanical properties of materials.<br>Prepare a short note on "Importance of Mechanical Properties in Chemical Plant Design".<br>Write an assignment covering CO-3 and CO4.<br>Preparation for Series II covering CO3 and CO4                                    | 6     |
| Total Self-Learning hours |                                                                                                                                                                                                                                                                                                                   | 90    |

### Learning Resources

| TEXT BOOK |                                      |                          |              |
|-----------|--------------------------------------|--------------------------|--------------|
| Sl. No.   | Title                                | Author                   | Publication  |
| 1         | Introduction to Chemical Engineering | S. Pushpavanam           | PHI Learning |
| 2         | Electrical Technology Vol-I          | B L Theraja, A K Theraja | S. Chand     |

|   |                                             |                          |          |
|---|---------------------------------------------|--------------------------|----------|
| 3 | Applied Mechanics and Strength of Materials | R. S. Khurmi & N. Khurmi | S. Chand |
|---|---------------------------------------------|--------------------------|----------|

| REFERENCE |                                      |                                |                  |
|-----------|--------------------------------------|--------------------------------|------------------|
| Sl. No.   | Title                                | Author                         | Publication      |
| 1         | Introduction to Chemical Engineering | S. Pushpavanam                 | PHI Learning     |
| 2         | Basic Electrical Engineering         | D. P. Kothari & I. J. Nagrath  | Tata McGraw-Hill |
| 3         | Engineering Mechanics                | Abhijit Chanda & Debabrata Nag | Wiley            |

| WEB RESOURCES |                                                                                                       |                 |
|---------------|-------------------------------------------------------------------------------------------------------|-----------------|
| Sl. No.       | URL                                                                                                   | Modules Covered |
| 1             | <a href="https://www.youtube.com/watch?v=i7t-MkFi6DU">https://www.youtube.com/watch?v=i7t-MkFi6DU</a> | I               |
| 2             | <a href="https://nptel.ac.in/courses/108105112">https://nptel.ac.in/courses/108105112</a>             | III             |

#### Assessment Methodology - Theory

| Mode               | Attendance      | CA1                                 | CA2                                 | CA3                                 | CA4                                 | CA5                     | CA6                     | ESE                   |
|--------------------|-----------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------|-------------------------|-----------------------|
|                    |                 | Self-learning activities (Module 1) | Self-learning activities (Module 2) | Self-learning activities (Module 3) | Self-learning activities (Module 4) | Series Exam (2 modules) | Series Exam (2 modules) | Written Exam (theory) |
| Duration           |                 |                                     |                                     |                                     |                                     | 2 hours                 | 2 hours                 | 2.5 hours             |
| Exam mark          |                 | 15                                  | 15                                  | 15                                  | 15                                  | 50                      | 50                      | 60                    |
| Converted to       |                 |                                     |                                     |                                     |                                     | 20                      | 20                      |                       |
| Marks              | 5               | 15                                  |                                     |                                     |                                     | 20                      |                         | 60                    |
| Total              | 40              |                                     |                                     |                                     |                                     |                         |                         | 60                    |
| Tentative schedule | End of semester | By 4th week                         | By 7th week                         | By 11th week                        | By 14th week                        | 8th week                | 15th week               | End of semester       |

#### Series Examination Pattern- Theory

| Time    | Module         | Part A | Marks | Part B | Marks | Part C (With choice) | Marks | Total |
|---------|----------------|--------|-------|--------|-------|----------------------|-------|-------|
| 2 hours | Any one module | 2      | 1     | 3      | 3     | 2                    | 7     | 25    |
|         | Any one module | 2      | 1     | 3      | 3     | 2                    | 7     | 25    |
|         | Number         | 4      |       | 6      |       | 4                    |       | 50    |

#### End Semester Examination Pattern of Question Paper - Theory

| Duration  | Module | TYPE OF QUESTIONS                        |                        |                                                  |                         |                                                          |                         | Total Marks |
|-----------|--------|------------------------------------------|------------------------|--------------------------------------------------|-------------------------|----------------------------------------------------------|-------------------------|-------------|
|           |        | Part A<br>(2 Questions from each module) |                        | Part B<br>2 out of 3 questions from each module) |                         | Part C<br>(1 set questions with choice from each module) |                         |             |
|           |        | No. of Questions                         | Marks<br>(1 mark each) | No. of Questions                                 | Marks<br>(3 marks each) | No. of Questions                                         | Marks<br>(7 marks each) |             |
| 2.5 hours | 1      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           | 2      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           | 3      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           | 4      | 2                                        | 2                      | 3                                                | 6                       | 1 set                                                    | 7                       | 15          |
|           |        | 8                                        | 8                      | 12                                               | 24                      | 4 sets                                                   | 28                      | 60          |